

Hajee Mohammad Danesh Science and Technology University, Dinajpur  
Department of Computer Science and Engineering  
M. Sc. in Computer Science and Engineering  
Semester Final Examination 2024 (Jan-June)  
Course Code: CSE 501, Credit: 3.0  
Course Title: Computer Arithmetic

Time: 3 Hour

Full Marks: 60

[N.B. The figure in the right margin indicates the marks for respective question and split answer to any question is unacceptable.]

Answer any 4 (Four) from the following questions

1. a) Define ALU. Assume an integer  $A$  composed of  $n$ -bit sequence of binary digits as  $a_{n-1}a_{n-2}\dots a_1a_0$ . Represent this integer  $A$  in sign magnitude and twos complement representation using generalized equation. 5
- b) Introduce the characteristics of twos complement representation and arithmetic. 5
- c) State and explain the overflow condition in twos complement addition with proper example. 5
2. a) Briefly introduce different types of floating-point formats defined by IEEE 754-2008. 4
- b) Make fair comparison between the representation of twos complement integer and floating-point numbers in 32-bit formats with proper example. 5
- c) Convert the following 32-bit floating-point number into decimal- 6
  - i) COB40000H, ii) 0 10000111 010010010110010000000000.
3. a) How does Booth's algorithm work for both positive and negative multiplier? 4
- b) Differentiate between traditional Booth's algorithm and modified Booth's algorithm. 4
- c) Explain the multiplication procedure of  $(-13) \times (-7)$  using modified Booth's algorithm. 7
4. a) Briefly introduce the approaches for division by constant. 5
- b) Calculate the instruction sequence for dividing the number  $z$  by 5 (Assume, 24 bits of precision,  $d = 5, m = 3$  and  $n = 4$ ). Count the total number of shift and addition operations. 5
- c) Explain the difficulties of quotient digit selection with example. 5
5. a) Explain the basics of floating-point algorithm for addition operation with proper example. 5
- b) Define logarithmic number system including its properties. 4
- c) Justify Euclid's algorithm for finding the GCD of two non-negative integers  $u$  and  $v$  with example. 6
6. a) Explain Radix-4 multiplication in dot notation with appropriate example. 5
- b) Introduce possible design for a Radix-4 multiplier. How does such design make some precomputation? 5
- c) Write integer addition algorithm and mention the meaning of all notations. 5

**Hajee Mohammad Danesh Science and Technology University, Dinajpur-5200**

**Department of Computer Science and Engineering**

**M.Sc. (Engg) in Computer Science and Engineering (CSE)**

**Semester Final Examination 2024 (Jan-Jun)**

**Course Title: Advanced Database Systems**

**Course Code: CSE 503      Credit: 3.0**

**Duration: 3 Hours**

**Total Marks: 60**

*[NB: The figure in the right margin indicates the marks for the respective question and split answer of any question is unacceptable]*

**Section-A**

**Answer any 6 (Six) questions from the following**

1. a) Define data abstraction. Briefly discuss about different levels of data abstraction 5  
b) Illustrate the need for a database system, giving suitable examples. 5
2. Create an ER diagram for each of the following descriptions: 5\*2
  - i. Each company operates four departments, and each department belongs to one company.
  - ii. Each department in part (i) employs one or more employees, and each employee works for one department.
  - iii. Each of the employees in part (ii) may or may not have one or more dependants, and each dependant belongs to one employee.
  - iv. Each employee in part (iii) may or may not have an employment history.
  - v. Represent all the ER diagrams described in (i), (ii), (iii), and (iv) as a single ER diagram.
3. a) Define normalization. Explain different types of normal forms using in database. 7  
b) Explain integrity constraints. 3
4. a) Define data warehousing and data mining. Describe a data warehouse architecture with relevant components. 5  
b) When a detection algorithm determines that a deadlock exists, the system must recover from the deadlock. Describe the actions need to be taken in that situation. 5
5. a) Why is B+ tree efficient than B tree? Explain how data retrieval, insertion and deletion are done using B tree and B+ tree indices. 5  
b) Describe the ACID properties of a Transaction. 5
6. a) Describe simple locking protocol. What problems may arise while using simple locking. 2+3  
b) Define transitive functional dependencies. Discuss the merits and demerits of normalization. 2+3

7.

**Customer table:**

| Customer_ID | Name  | City        |
|-------------|-------|-------------|
| 1           | Mike  | New York    |
| 2           | Alice | Los Angeles |
| 3           | Sara  | Chicago     |
| 4           | Jane  | New York    |

**Orders Table:**

| Order_ID | Customer_ID | Order Date | Order total |
|----------|-------------|------------|-------------|
| 100      | 1           | 2023-07-01 | 100.00      |
| 101      | 2           | 2023-06-15 | 50.00       |
| 102      | 3           | 2023-07-05 | 150.00      |
| 103      | 1           | 2023-07-07 | 75.00       |
| 104      | 4           | 2023-07-02 | 200.00      |

Write the SQL statements for the following:

- Find the total number of orders placed by each customer, excluding orders placed in June.
- Find the customer who has placed the highest total order value.
- List all orders placed on specific dates (eg., 2023-07-04 and 2023-07-06) and their corresponding customer names.
- Find the average order value for each city.
- Identify customers who haven't placed any orders.

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- How many types of database-system architecture are there? Mention them.
- Explain distributed system. Discuss about the distributed database system with figure.

3

1+6

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Semester Final Examination 2024 (Jan-Jun)  
Course Code: CSE 505, Credit: 2.0  
Course Title: Research Methodology

Time: 2 Hours

Marks: 40

Answer any Four

1. a) Write the possible steps of a research. Explain any two of them. 2+4  
b) "No contribution to new knowledge is not a research". Discuss from your point of view. 4
2. a) Compare research methodology with research method with example. 6  
b) Write the tentative title of your M.Sc. research. What type of research it is and why? 4
3. a) Write the available research design approaches in Computer Science. Which one among them you would following your M.Sc. research and why? 2+4  
b) Define computer simulation with example. 4
4. a) What is research ethics? How to avoid misconduct in scientific research? 2+4  
b) What are the possible ways to follow to avoid/prevent plagiarism in research? 4
5. a) How a survey based research is conducted? 6  
b) Differentiate between primary and secondary research data. 4
6. Write a short note on any of the following: 10  
i. Bibliography/Reference ii. Research proposal

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Semester Final Examination 2024 (Jan-June)

Course Code: CSE 517, Credit: 2.0

Course Title: Software Testing and Quality Assurance

Time: 2 Hours

Total Marks: 40

*[N.B. The figure in the right margin indicates the marks allocated for respective question.  
Split answer of any question is not allowed.]*

*(Answer any 04 (Four) from the following questions)*

1. a) Define software testing. Briefly discuss about the goals and objectives of software testing. 1+4
- b) Differentiate between software quality assurance and software quality control. 5
2. a) What is the importance of acceptance testing? 2
- b) Briefly introduce the test process in software testing. 8
3. a) Illustrate Agile model of software testing. 4
- b) Explain boundary value analysis testing technique with suitable example. How does it differ from equivalence partitioning? 6
4. a) Define integration testing. State and describe the approaches of integration testing. 1+4
- b) Write a short note on code inspection. 5
5. a) Differentiate among errors, faults and failures. 2
- b) Describe the following: 8
  - i) Load testing
  - ii) Recovery testing
  - iii) Beta testing
  - iv) Vulnerability testing
6. Discuss in detail about defect/bug life cycle in software testing. 10

Time: 2 Hours

[N.B. The figure in the right margin indicates the marks allocated for the respective question.  
Split answer to any question is not allowed.]

Answer any 4 (four) from the following questions. Question 1 is mandatory to answer.

1. a) What is Data Mining? Briefly discuss the steps involved in the data mining process. 5  
b) Define machine learning. How do supervised and unsupervised learning differ fundamentally? 5

2. a) What are the primary differences between machine learning, data mining, and pattern recognition? 5  
b) Find out the value of Accuracy, Precision, Recall, F1 – score, FN rate, TN rate from the following confusion matrix: 5

|             |               |                |
|-------------|---------------|----------------|
| N = 193     | Predicted: NO | Predicted: YES |
| Actual: NO  | 75            | 10             |
| Actual: YES | 8             | 100            |

3. a) What do you understand by association rule mining? Write short notes on the Apriori algorithm. 5  
b) From the given table below, find frequent item sets and generate the association rule using the Apriori algorithm: (maximum support = 2, minimum confidence = 50%) 5

| TID | ITEM SETS  |
|-----|------------|
| T1  | A, B       |
| T2  | B, D       |
| T3  | B, C       |
| T4  | A, B, D    |
| T5  | A, C       |
| T6  | B, C, D    |
| T7  | A, C       |
| T8  | A, B, C, E |
| T9  | A, B, C    |

4. Define the curse of dimensionality problem. Given data points {2, 1, 4, 5, 6, 7; 1, 5, 3, 6, 7, 2} 10  
I. Compute the principle component using the PCA Algorithm.  
II. Use the PCA algorithm to transform the pattern (2, 2) onto the eigenvector in the previous question.

5. a) What is clustering? Write a short note on the K-means algorithm. 5  
b) Suppose that the data mining task is to cluster points (with (x, y) representing location) into three clusters, 5  
where points are A1(2,10), A2(2,5), A3(8,4), B1(5,8), B2(7,5), B3(6,4), C1(1,2), C2(4,9) and the distance function is the Euclidean distance. Initially, you should assign A1, B1, and C1 as the center of each cluster, respectively. Now, use the k-means algorithm to show only  
i. The three cluster centers after the first round of execution.  
ii. The final three clusters.

6. Consider a data set where data points (2,2), (2,-2), (-2,-2), (-2,2) are labelled positively and data points (1,1), (1,-1), (-1,-1), (-1,1) are labelled negatively. Find the Hyper-plane coordinates with bias and weight. Where kernel SVM condition: 10

$$\theta_1 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} > 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

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Class Test Examination 2024 (Jan-June)  
Course Code: CSE 501, Course Title: Computer Arithmetic

Time: 80 Minutes

Full Marks: 60

Answer any 6 (Six) from the following questions

1. a) Sketch the block diagram of hardware for addition and subtraction operation. 4  
b) Compute the product ( $\text{multiplier} = 1010_b$  and  $\text{multiplicand} = 0101_b$ ) by using unsigned binary multiplication. 6
2. a) Explain all the notations with example in  $\pm S \times B^{\pm E}$ . 5  
b) Mention the features of typical 32-bit floating point number. 5
3. a) What are the benefits using the Booth's algorithm than the straightforward approach? 3  
b) Introduce Booth's algorithm. Explain the multiplication procedure of  $(-7) \times (-5)$  by using the Booth's algorithm. 7
4. a) Compute  $29 \div 5$  (for  $z = 29$  and  $d = 5$ ). Write the comments of each step. 5  
b) Sketch the approach for division of an 8-bit number by a 4-bit number in dot notation. 5
5. a) Write about the advantages of Logarithmic representation of numbers. 5  
b) Write the algorithm for calculation square root of integer. 5
6. a) Which approach is preferred in radix-4 multiplication? Briefly explain it. 5  
b) Sketch the block diagram of hardware implementation for unsigned binary multiplication. 5
7. a) Write short notes on requirements for arithmetic. 5  
b) Mention the properties of logarithmic representation. 5

### Class Test

Course Code: CSE 517

Time: 40 Minutes

1. Differentiate between the following: 10
  - i) Testing and Debugging
  - ii) Black box testing and White box testing
2. State the characteristics of a good test case. 5
3. Write a short note on: 10
  - i) Test harness
  - ii) Unit testing
4. Define the following: 5
  - i) Monkey Testing
  - ii) Regression Testing